

SMC CPU Cluster — Specification

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SMC CPU Cluster — Specification

Document: 01_CPU_Cluster_Specification.md
Module: smc::smc_cpu_cluster
Status: Active — SystemC / TLM-2.0 Loosely-Timed model

1. Purpose

The SMC CPU Cluster is the per-chiplet **RV64GC management processor** (1–4 harts) in the Open Chiplet Atlas SMC. This SystemC model wraps the Tenstorrent **Whisper** instruction-set simulator behind an `iss_hart` interface so the rest of the SMC IP library can bind TLM sockets and IRQ wires without depending on Whisper headers.

Normative hardware references:

Document	Relevance
01_SMC_Architecture.pdf §3	IP role, parameters, chiplet context
02_SMC_IP_LowLevel_Design.pdf §3	TLM sockets, register map, IRQ, routing
03_SMC_Test_Plan.pdf §A.1–A.6, §B.3	Project-wide verification tiers

Implementation detail: 02_CPU_Cluster_LowLevel_Design.md.
Verification detail: 03_CPU_Cluster_Test_Plan.md.

2. Module boundary

```
SC_MODULE(smc_cpu_cluster) {  
    // Initiators (§3.6 bus-bridge routing)  
    tlm_utils::simple_initiator_socket<smc_cpu_cluster, 64> data;  
    tlm_utils::simple_initiator_socket<smc_cpu_cluster, 64> mmio;  
    tlm_utils::simple_initiator_socket<smc_cpu_cluster, 64> ifetch;  
  
    // CPU-Control register file target (§3.8, 8 KiB window)  
    tlm_utils::simple_target_socket<smc_cpu_cluster, 64> ctrl;  
  
    // Per-hart IRQ inputs (§3.7)  
    sc_vector<sc_in<bool>> irq_sw;      // CLINT MSIP  
    sc_vector<sc_in<bool>> irq_timer;   // CLINT MTIP  
    sc_vector<sc_in<bool>> irq_ext;     // PLIC MEIP  
  
    explicit smc_cpu_cluster(sc_module_name name);  
    smc_cpu_cluster(sc_module_name name, const config& cfg); // legacy struct  
};
```

Every outgoing transaction carries `smc_axi_extension` (source ID, hart ID, privilege, AMO lock hint).

3. Configuration

Runtime parameters are exposed as **SystemC CCI 1.0** `cci_param` members (broker presets override defaults before construction). The legacy `smc_cpu_cluster::config` struct remains for backward-compatible construction.

Parameter	Default	Range / note
<code>num_harts</code>	1	1..4
<code>reset_pc</code>	<code>0x80000000</code>	Per-hart boot vector
<code>isa</code>	<code>rv64imafdc</code>	Whisper ISA string
<code>fast_mem_lo / fast_mem_hi</code>	<code>0 / 0x80000000</code>	Internal flat memory window
<code>mmio_lo / mmio_hi</code>	<code>0x80000000 / 0x90000000</code>	MMIO carve-out (§3.6)
<code>quantum_ns</code>	1000	TLM LT global quantum (~1 µs)
<code>quantum_insts</code>	1000	Instructions per <code>step(K)</code> slice
<code>amo_lock_detect</code>	true	Assert <code>prot[3]</code> on AMO / LR-SC
<code>source_id</code>	<code>0x10</code>	<code>SMC_CPU_SOURCE_ID</code> on extension
<code>ctrl_size_bytes</code>	<code>0x2000</code>	CPU-Control aperture size
<code>local_base_default</code>	<code>0xC0000000</code>	RO <code>LOCAL_BASE</code> reset value

See `04_CCI_Integration_Guide.md` for broker setup and migration notes.

4. Functional requirements (summary)

ID	Requirement	Spec ref
F1	1..4 <code>rv64imafdc</code> harts under TLM LT quantum	§3.4–3.5
F2	Fast-mem buffer + MMIO / data socket routing	§3.6
F3	CPU-Control CSR target at <code>BASE+0x001_0000</code>	§3.8
F4	IRQ aggregator maps MSIP/MTIP/MEIP → MIP	§3.7
F5	WFI park / IRQ wake per hart	§3.7
F6	<code>smc_axi_extension</code> on all initiator traffic	§3.10
F7	Whisper ISS via memory callbacks (<code>MEM_CALLBACKS=1</code>)	§A ISS integration
F8	Debug API: <code>load_elf</code> , <code>hart(i)</code> , <code>inject_nmi</code> , CSR peek	§3.9

Full traceability matrix: `03_CPU_Cluster_Test_Plan.md` §7.

5. Out of scope

- Full PLIC IP verification (owned by `peripherals/plic/plic_tb`)

- CLINT MMIO model (IRQ wires are stubbed in `cluster_tb`)
 - Production firmware / SoC-level tests
 - Spike or non-Whisper ISS backends
-

6. Build & run

```
cd smc/cpu_cluster
cp deps.env.example deps.env # first time - edit paths
./run_tests.sh
```

Dependencies: SystemC (C++20), Boost, Whisper source (auto-built by script), SystemC CCI 1.0. See `README.md` for first-time setup.

Coverage gate: $\geq 95\%$ line coverage on `src/` via `./run_tests.sh --coverage` (current baseline **97%** — see test plan §9.2).